1. Two liquids A and B are in the ratio 5:1 in container 1 and in the ratio 1:3 in container 2. In what ratio should the contents of the two containers be mixed so as to obtain a mixture of A and B in the ratio 1:1?
(Mixture) NIMCET-2019
(a)2:3 (b) 4:3 (c)3:2 (d)3:4

2. A 30L mixture of milk and water contains 10% water. How much milk should be added so that the percentage of water in the mixture comes down to 2%?
(Mixture) NIMCET-2023
(a)120L (b)80L (c)90L (d)60L

3. A vessel contains total 95L mixture of milk and water in the ratio of 15: 4 respectively. P L of mixture taken out from the vessel and 18L water added in remaining mixture, then the new ratio of milk to water becomes 3: 2, find the value of P.
(Mixture) NIMCET-2023
(a)57 (b)19 (c)27. 5 (d)38

ANSWER

1	d	2	a	3	d		

SOLUTION:

1. (d) Let the required ratio be x : y.

Liquid A in x L of mixture in container 1 = $\frac{5x}{6}$ L

Liquid B in x L of mixture in container 1 = $\frac{x}{6}$ L

Liquid in y L of mixture in container 2 = $\frac{y}{4}$ L

Liquid B in y L of mixture in container 2 = $\frac{3y}{4}$ L

According to the question,

$$\left[\frac{5x}{6}+\frac{y}{4}\right]:\left[\frac{x}{6}+\frac{3y}{4}\right]=1:1$$

Given, $\frac{\frac{5x}{6}+\frac{y}{4}}{\frac{x}{6}+\frac{3y}{4}}=\frac{1}{1}$

$\Rightarrow \frac{5x}{6}+\frac{y}{4}=\frac{x}{6}+\frac{3y}{4}$

$\Rightarrow \frac{5x}{6}-\frac{x}{6}=\frac{3y}{y}-\frac{y}{4}$

$\Rightarrow \frac{4x}{6}=\frac{2y}{4}$

$\Rightarrow \frac{2x}{3}=\frac{y}{2}$

$\Rightarrow \frac{x}{y}=\frac{3}{4}$

64 1 x3y 1 5x + 6 4 6 4 5x 6 y 4x2y 6 3 123-4 4 y

Hence, the required ratio is 3:4.

2. **(a)** In 30 L of mixture,

Water = 10% of 30 L = 3L

Let milk to be added = xL

$\frac{3}{30+x}\times 100=2 \Rightarrow 60+2x=300$

$\Rightarrow$ x = $\frac{300-60}{2}$ = 120 L

3. **(d)** Total milk in mixture initially

$= 95\times\frac{15}{19}=75L$

Total water in mixture initially

$= 95\times\frac{4}{19}=20L$

According to the question,

$\frac{75-P\times\frac{15}{19}}{20-P\times\frac{4}{19}+18}=\frac{3}{2}$

So P = 38

Hence, 38L of mixtures was taken out from the vessel.